QDB SME Relief Portal

An emergency financing portal that enables Qatari SMEs affected by geopolitical disruption to apply for relief under the National Relief and Guarantee Program (NRGP), with automated eligibility checking, WPS-validated salary verification, and direct routing into QDB's Dynamics CRM.

Business Context

The Problem

Qatar's SME sector is under acute cash flow stress from the Iran-Israel regional conflict. Trade routes are disrupted, supply chains are fragmented, and regional demand is suppressed. Affected SMEs face simultaneous fixed obligations: salary payments legally mandated under Qatar's Wage Protection System (WPS) and commercial lease obligations.

Without emergency financing, these businesses face:

- WPS violations — QAR 6,000 per employee per late salary cycle (QAR 300,000 for a 50-person SME)
- Lease defaults and potential eviction from commercial premises
- Staff layoffs and business closure

The Program

The National Relief and Guarantee Program (NRGP) is an existing QDB financing instrument with board-level approval, an active beneficiary list from prior activations (COVID-19), and pre-negotiated guarantee structures. QDB is not creating a new financial product — it is digitalising delivery of an approved, standing program under an emergency activation mandate.

The Portal's Purpose

A structured digital workflow replaces a manual, paper-heavy process that takes 15-30 business days to process a single application. The portal:

1. Authenticates applicants via Tawtheeq (Qatar's National Authentication Service / NAS)
2. Retrieves verified company data from the Ministry of Commerce and Industry (MOCI)
3. Evaluates eligibility automatically against NRGP rules
4. Validates salary obligations against WPS payroll records
5. Routes disbursement requests into Microsoft Dynamics CRM — automatically for returning NRGP beneficiaries, or for manual QDB review for new applicants

Urgency: A 6-8 week delivery target is required. Every month of delay exposes SMEs to WPS violations and irreversible business harm.

Who It Serves

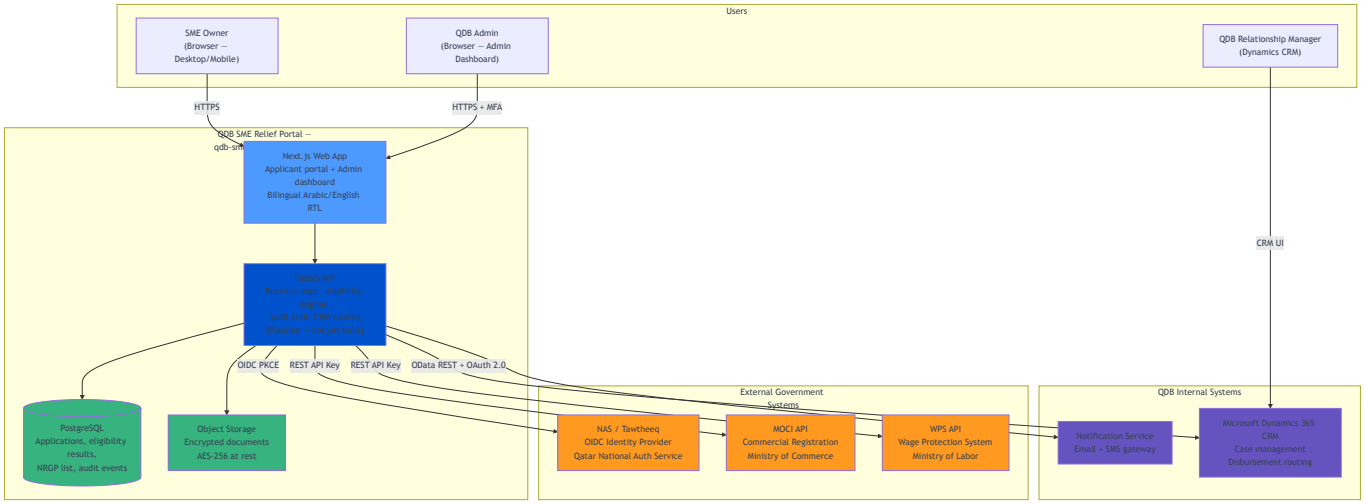
Persona	Role	Primary Need
Khalid (SME Owner)	Applies for relief on behalf of his company	Fast, Arabic-language portal that confirms eligibility upfront and guides document submission
Fatima (QDB RM)	Reviews manual disbursement cases in Dynamics CRM	CRM cases that arrive pre-populated with all data, correctly routed, WPS-flagged
Mohammed (QDB Admin)	Manages NRGP list, eligibility rules, program lifecycle	Admin interface with full control over program parameters without engineering

Success Metrics

Metric	Target
Auto-disbursement end-to-end time	< 1 business day
Manual review processing time	< 5 business days
Document re-submission rate	< 10% (down from 40-60% manual)
Applications processed per week	500+ (up from ~20 manual)
Portal uptime during program window	> 99.5%
Duplicate application detection	100%
Audit trail completeness	100% of steps logged

Architecture Overview

The portal is delivered as a Next.js web application (current prototype stage), with a planned Fastify API service, PostgreSQL database, object storage, and Dynamics CRM integration.



Quick Start

Prerequisites

- Node.js 20+
- pnpm 9+

Run the Prototype (Port 3120)

The current implementation is a Next.js prototype (frontend only — no backend API yet).

```
# From the product root
cd products/qdb-sme-relief/apps/web

# Install dependencies
pnpm install

# Start the development server
pnpm dev
```

The portal will be available at `http://localhost:3120`.

What the Prototype Includes

The prototype demonstrates the full applicant journey as a UI walkthrough:

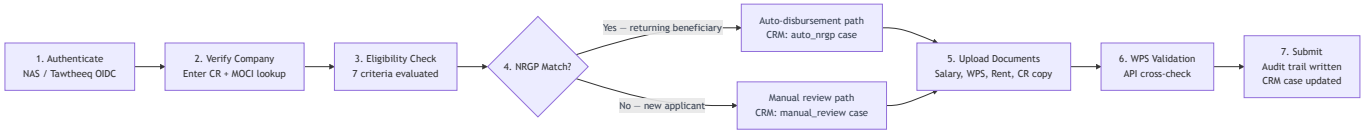
- Language selection (Arabic / English with RTL toggle)
- NAS/Tawtheeq authentication flow (mocked)
- CR number entry and MOCI company data display (mocked)
- Eligibility evaluation result screen
- NRGP list check and CRM routing decision

- Document upload UI (all document types)
- WPS validation result screen
- Application submission confirmation and status tracking

> **Note**: All external API calls (NAS, MOCI, WPS, CRM) are mocked in the prototype. Production integration requires a live Fastify API service (see docs/ARCHITECTURE.md).

Application Flow

The portal guides SMEs through a structured seven-step journey from authentication to submission.



Key Integrations

NAS / Tawtheeq (Authentication)

- Protocol: OpenID Connect (OIDC) with PKCE
- Purpose: Authenticate applicants using their existing Qatar national digital identity
- No user passwords stored in the portal — NAS is the sole identity source
- QID claim extracted from the NAS ID Token is used for all subsequent signatory checks

MOCI (Company Registration)

- Protocol: REST API with API key authentication
- Purpose: Retrieve verified company data (name, CR status, sector, registration date, signatories)
- CR number entered by the applicant is the lookup key
- Signatory list cross-referenced against the authenticated QID to confirm authorization

WPS (Wage Protection System)

- Protocol: REST API with API key authentication
- Purpose: Validate declared salary obligations against Ministry of Labor payroll records
- Discrepancies exceeding 10% flag the CRM case for manual review (not a block)
- Applicants may also upload a WPS CSV file as evidence

Microsoft Dynamics CRM

- Protocol: OData REST API with Azure AD service principal (OAuth 2.0)

- Purpose: Create case records routed to QDB operations for disbursement processing
- Two case types: `auto_nrgp` (pre-vetted, fast path) and `manual_review` (new applicant)
- Portal polls CRM every 5 minutes to sync application status back to the applicant

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Prototype vs Production Scope

Capability	Prototype (Current)	Production (Planned)
UI / UX flows	Complete	Complete + refined from testing
NAS / Tawtheeq auth	Mocked	Live OIDC integration
MOCI CR lookup	Mocked	Live REST API
WPS validation	Mocked	Live API + file parser
Dynamics CRM routing	Mocked	Live OData REST integration
Document upload	UI only	S3-compatible object storage + virus scan
Eligibility engine	Frontend simulation	Rule-engine in Fastify API, DB-backed config
Audit trail	None	Append-only PostgreSQL table
Bilingual / RTL	Implemented	Professionally translated Arabic strings
Admin dashboard	Not started	Full NRGP list mgmt, eligibility rule config
Notifications	None	Email + SMS via gateway
Program lifecycle	None	Admin-controlled open/paused/closed states

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Eligibility Criteria

An SME must pass all seven criteria to proceed:

Criterion	Rule
EC-001	CR status is "active" in MOCI
EC-002	Company registered for at least 12 months
EC-003	Meets QDB SME definition (employees or revenue threshold)
EC-004	Enrolled in WPS (employee count > 0)
EC-005	Operates in an impacted sector OR declares a measurable revenue decline
EC-006	No active QDB loan in default
EC-007	No judicial dissolution order in MOCI records

All criteria parameters are admin-configurable without a code deployment (production).

Security and Compliance

- All data in transit secured with TLS 1.3 minimum
- Documents encrypted at rest using AES-256 with KMS-managed keys
- Document access via signed, time-limited URLs (1-hour expiry)
- All uploaded files virus-scanned before activation
- Audit trail is append-only — UPDATE and DELETE are prevented at the database constraint level
- Documents retained for 7 years minimum per QDB policy and Qatar PDPA requirements
- Admin access requires multi-factor authentication
- Rate limiting: maximum 3 application attempts per CR number per 24 hours

Full Documentation

Document	Description
docs/PRD.md	Full Product Requirements Document with all user stories and acceptance criteria
docs/ARCHITECTURE.md	C4 architecture diagrams, sequence diagrams, ER diagram, security architecture
docs/API.md	API reference for the planned Fastify backend
docs/USER-GUIDE.md	End-user guide for SME applicants
docs/ADMIN-GUIDE.md	QDB admin operations guide
docs/business-analysis.md	Business analysis, feasibility assessment, risk register
docs/ADRs/	Architecture Decision Records